MATH118 Probability and Statistics 2025 Fall Syllabus

Catalog Description

This course introduces the principles of probability and statistics with engineering applications. Topics include probability laws, random variables, common discrete and continuous distributions, expectation, joint and conditional distributions, the Central Limit Theorem, confidence intervals, hypothesis testing, and regression analysis. The course also introduces Bayesian inference, emphasizing probabilistic modeling and decision-making under uncertainty. Statistical software is used for data analysis, simulation, and model interpretation.

Educational Objectives

On completion of this course, you should be:

- Able to understand and apply foundational concepts in probability, including random variables, distributions, and expected value.
- Competent in using both frequentist and Bayesian approaches to statistical inference in engineering contexts.
- Skilled at formulating and solving real-world problems involving uncertainty using appropriate probabilistic models.
- Proficient in analyzing data with methods such as hypothesis testing, confidence intervals, regression, and posterior inference.
- Capable of using statistical software to simulate probabilistic models and analyze empirical data.
- Prepared to critically evaluate modeling assumptions and assess the reliability of data-driven decisions in engineering systems.

Instruction

Meeting Times and Places

- There won't be a live/recorded online lecture.

Section 1

- **Lecture:** Tuesday 09:30-10:30 Z10
- **Lecture:** Thursday 10:30-12:30 Z23

Section 2

- **Lecture:** Tuesday 16:30-17:30 Z23
- **Lecture:** Thursday 13:30-15:30 Z23

Instructor

- Gokhan Kaya
  - Email: gokhankaya@gtu.edu.tr
  - Room: 218
  - Office Hours: Wednesday, 10:30 - 12:30

Teaching Assistants

- Graders:
  - Arif Burak Dikmen, Sümeyye Ünsür, Şeydanur Ahi

Management

- This course will be managed through a Microsoft Teams team. You will be added to the course's Teams page after the first week.
- You are responsible for all announcements made in class, as well as any videos, documents, or materials published on the course's Teams page. This includes all content covered in online lectures and any information shared via Microsoft Teams.

Communication

- You may contact the instructor via email.
- You must use your official university email address; messages from personal or unofficial accounts will not be responded to.
- Please begin the subject line of your email with: MATH118F25
- If you need assistance, you may either schedule an online meeting or visit during office hours.

Textbooks

- Probability of Statistics for Engineering & Scientists. Walpole E.W., Myers R.H., Myers S.L., Ye K. Pearson Education, Prentice Hall Inc.
- Probability and Statistics for Engineering and the Sciences, Jay L. Devore
- Probability Random Variables and Stochastic Processes, A.Papolis, McGraw Hill

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Grading (Tentative)

- Items:
  - Midterm Exams:
    - 1 count (Date: 7^th or 8^th week of the semester)
    - in-class written exam
    - weight: 35%
  - Final Exam:
    - 1 count (Date: will be announced later)
    - in-class written exam
    - weight: 35%
  - Homework Assignments and Quizes:
    - 5 counts
    - individual homeworks
    - total weight: 30%
    - After every homework or assignment, there will be an in-class quiz. The questions will be similar to the homework and they will be about the same content.
    - If you don't attend the related quiz and get a passing grade(50 out of 100), your homework or assignment will not be graded.
    - Homework and quiz passing grade is 50 out of 100.
- Letter Grades:

Score Range	Letter Grade
91-100	AA
85-90	BA
80-84	BB
70-79	CB
65-69	CC
55-64	DC
40-54	DD
39 and below	FF

HW and Quiz Grading Policy

- There will be 5 homework assignments and 5 quizzes. Each homework will be followed by a quiz, and the content of each pair will be related.
- Homeworks must be handwritten, scanned using a smartphone app, and submitted as a PDF document. Submissions in other formats will not be graded.

- Each homework and quiz pair will be graded together, and you will receive a single grade for each pair.
- If you do not submit the homework, or if your submission is irrelevant, a dummy, or shows no effort, your grade for that pair will be 0. (The homework will be considered not submitted.)
- If you do not attend the quiz, your grade for that pair will be 0.
- If you submit a partial answer for the homework, you will receive a fraction of your quiz grade for the pair.
- All class-related announcements will be made either in class or on the class team page.
- Students are required to read their emails regularly and check the team page.
- Homework assignments will be announced and submitted via the team page.
- If you have an official excuse and miss a submission, you must inform the instructor within a week and present the related official documents.
- It is the student's responsibility to foresee possible technical problems that may occur while submitting the assignment. In case of a failure to submit just before the deadline, no second chance will be given. Late submissions will not be accepted unless stated otherwise.

NA and VF Conditions

- A student must submit and pass at least 3 homework assignments **and** 3 quizzes. Failure to do so will result in a VF (Fail due to withdrawal or inactivity) grade.
- A student who does not meet the attendance requirement will receive an NA (Not Attending) grade.
- A student who does not take the midterm exam will also receive a VF grade.
- Dishonest behavior regarding attendance (e.g., signing in for others) may lead to an NA grade.
- Students who receive either VF or NA will not be allowed to take the final exam.
- If a student does not attend the final exam, the final grade will be FF, or VF/NA depending on which condition(s) apply.

D-Exam

- D-Exam passing grade is 69 out of 100.

Attendance (Tentative)

- Attendance will be recorded and may impact your final grade. You are expected to attend at least 70% of the lectures.
- Please review the Attendance section of the official university regulations, as it may be subject to updates.
- While attendance is strongly recommended, it may not be mandatory depending on the current university policy.
- However, if attendance is required and you fail to attend at least 70% of the scheduled classes, you will receive an NA (Not Attending) grade.

Cheating

Unless otherwise specified, all work submitted for a grade—such as homework assignments and exams—must be the student's own individual effort. While discussing homework concepts with classmates is encouraged, this must not extend to sharing or co-writing detailed answers. Submissions that are identical beyond coincidence will be treated as academic misconduct. Disciplinary actions may include, but are not limited to, a failing grade for the assignment for all students involved. Each student is responsible for safeguarding their own work, whether in electronic or printed form.

Academic Honesty Rules: - Copying someone else's work—even with superficial changes like renaming variables or rearranging content—is considered cheating. - Cheating is strictly prohibited. - Do not cheat. All incidents will be officially reported to the Dean's Office. - Do not sign the attendance sheet if you are not physically present in the class. Detection will result in an NA (Not Attending) grade. - Do not simulate attendance in online lectures unless you are genuinely participating. Misrepresentation will also lead to an NA. - Any behavior violating academic integrity will be formally reported to the Dean's Office.

Weekly Schedule (Tentative)

Week & Dates	1-Hour Session (First)	2-Hour Session (Later)	HW / Quiz / Exam
Week 1 (22.09.2025 – 26.09.2025)	Course Introduction, grading, academic honesty	Why probability matters, set theory, sample space, events	Topics covered in **HW 1**
Week 2 (29.09.2025 – 03.10.2025)	Complements, Venn diagrams, axioms of probability	Counting techniques, conditional probability, independence	Topics covered in **HW 1**
Week 3 (06.10.2025 – 10.10.2025)	Guided problem solving: conditional probability, Bayes' theorem	Discrete random variables: PMF, expectation, variance	**HW 1 Assigned** (Week 1+2 topics, due Week 4, 1-hr session)
Week 4 (13.10.2025 – 17.10.2025)	**HW 1 Due + Quiz 1**	Continuous random variables: PDF, CDF, expectation	**Quiz 1** – Tests Week 1+2 topics
Week 5 (20.10.2025 – 24.10.2025)	Normal distribution, standardization, z-tables	Binomial, normal approximation	**HW 2 Assigned** (Week 3+4 topics, due Week 6, 1-hr session)
Week 6 (27.10.2025 – 31.10.2025)	**HW 2 Due + Quiz 2**	Joint distributions: discrete & continuous, marginal/conditional distributions	**Quiz 2** – Tests Week 3+4 topics
Week 7 (03.11.2025 – 07.11.2025)	Covariance, correlation, functions of RVs	Review session for Midterm	**HW 3 Assigned** (Week 5+6 topics, due Week 8, 1-hr session)
Week 8 (10.11.2025 – 14.11.2025)	**HW 3 Due + Quiz 3**	**Midterm Exam** (2-hour in-class)	**Quiz 3** – Tests Week 5+6 topics; Midterm Exam (2-hr)
Week 9 (17.11.2025 – 21.11.2025)	Point estimation: bias, MSE	Confidence intervals for population mean	Topics covered in **HW 4**
Week 10 (24.11.2025 – 28.11.2025)	Hypothesis testing: framework, Type I/II errors	One-sample and two-sample tests	
Week 11 (01.12.2025 – 05.12.2025)	**HW 4 Assigned** (Week 9+10 topics, due Week 12, 1-hr session)	Two-sample tests, p-values, power	
Week 12 (08.12.2025 – 12.12.2025)	**HW 4 Due + Quiz 4**	Linear regression, least squares estimation	**Quiz 4** – Tests Week 9+10 topics
Week 13 (15.12.2025 – 19.12.2025)	**HW 5 Assigned** (Week 11+12 topics, due Week 14, 1-hr session)	Practice/examples: Bayesian estimation	
Week 14 (22.12.2025 – 26.12.2025)	**HW 5 Due + Quiz 5**	Final exam prep / Optional topics: ANOVA, reliability, or review	**Quiz 5** – Tests Week 11+12 topics; Final Exam Preparation

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